B2.3 Cell specialization (HL)

NAME	DATE	TOTAL
		_____ / 30

About 30 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 STATE two ways in which the biconcave shape of a red blood cell helps it carry oxygen. [B2.3.7] [2]

2 NAME the vesicles in type II pneumocytes that store surfactant. [B2.3.8] [1]

3 STATE the role of intercalated discs in cardiac muscle. [B2.3.9] [1]

4 STATE the function of the mid-piece of a sperm cell. [B2.3.10] [1]

5 STATE two roles of the zona pellucida. [B2.3.10] [2]

Section B: Understanding

11 MARKS

6 EXPLAIN how cells of the proximal convoluted tubule are adapted for reabsorption. [B2.3.7] [3]

7 EXPLAIN why the alveolar epithelium needs two types of cell. [B2.3.8] [2]

8 DISCUSS whether a striated muscle fibre should be called a cell. [B2.3.9] [3]

9 **EXPLAIN** why a human egg is much larger than a sperm cell. [B2.3.10] [3]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [B2.3.8]

The table shows the percentage of premature babies who developed respiratory distress, by length of pregnancy. The L/S ratio is a measure of surfactant phospholipid in the fluid around the baby; a ratio of 2.0 or more usually means the lungs are mature.

LENGTH OF PREGNANCY / WEEKS	26	28	30	32	34	36
BABIES WITH RESPIRATORY DISTRESS / %	92	76	55	30	12	4
MEAN L/S RATIO	0.8	1.1	1.4	1.8	2.2	2.6

a **DESCRIBE** the relationship between length of pregnancy and respiratory distress. [1]

b **CALCULATE** the percentage decrease in respiratory distress between 28 and 34 weeks. [2]

c **EXPLAIN** why babies born very early often have difficulty breathing. [2]

d **DEDUCE** the earliest length of pregnancy in the table at which the lungs are usually mature. [1]

e Mothers likely to give birth early are often given a steroid hormone. **SUGGEST** how this could reduce respiratory distress in the baby. [2]

Section D: Longer answer

4 MARKS

11 **COMPARE** heart muscle cells with skeletal muscle fibres. [B2.3.9] [4]